Lecture 21:

- Differential Equations $\frac{du}{dt} = Au$
- Markov matrices
- Steady states

Example:

$$\frac{du_1}{dt} = -u_1 + 2u_2 \qquad u(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\frac{du_2}{dt} = u_1 - 2u_2$$

$$\longrightarrow A = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \qquad \lambda = 0, -3$$

$$\det(A - \lambda I) = \begin{vmatrix} -1-\lambda & 2 \\ 1 & -2-\lambda \end{vmatrix} = \lambda^2 - 3\lambda = \lambda(\lambda+3)$$

$\underline{\lambda_1 = 0}$ $\quad x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$\underline{\lambda_2 = -3}$ $\quad A + 3I = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Solution: $u(t) = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t} x_2$

Check: $\frac{du}{dt} = Au$

$$\lambda_1 \cancel{e^{\lambda_1 t}} x_1 \overset{\checkmark}{=} A \cancel{e^{\lambda_1 t}} x_1$$

(For $u_{k+1} = Au_k$, solution was $u_k = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2$)

$$u(t) = c_1 \cdot 1 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \cdot e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

At $t=0$: $\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\Rightarrow \quad c_1 = \frac{1}{3}, \quad c_2 = \frac{1}{3}$$

$$u(t) \longrightarrow \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix} \quad \text{as } t \to \infty.$$

Markov Matrices

$$A = \begin{bmatrix} .1 & .01 & .3 \\ .2 & .99 & .3 \\ .7 & 0 & .4 \end{bmatrix}$$

1. All entries ≥ 0
2. Columns add to 1.

- λ_1 is an eigenvalue, $x_1 \geq 0$
- All other $|\lambda_i| \leq 1$
- All other $|\lambda_i| < 1$ if entries > 0.

$$\Rightarrow u_k = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 + \cdots + c_n \lambda_n^k$$

$$\underset{k \to \infty}{\longrightarrow} c_1 x_1 \text{ (steady state)}$$

$$(A - 1 \cdot I)x_1 = \begin{bmatrix} -.9 & .01 & .3 \\ .2 & -.01 & .3 \\ .7 & 0 & -.6 \end{bmatrix} \begin{bmatrix} .6 \\ 33 \\ .7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Rows of A-I add to zero

$\Rightarrow$ A-I singular

$u_{k+1} = Au_k$ A Markov

$$\begin{bmatrix} u_{wash} \\ u_{cal} \end{bmatrix}_{k+1} = \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix} \begin{bmatrix} u_{wash} \\ u_{cal} \end{bmatrix}_k$$

$$\begin{bmatrix} u_{wash} \\ u_{cal} \end{bmatrix}_0 = \begin{bmatrix} 10 \\ 40 \end{bmatrix} \quad \begin{bmatrix} u_{wash} \\ u_{cal} \end{bmatrix}_1 = \begin{bmatrix} 17 \\ 33 \end{bmatrix} \quad \ldots$$

$$A = \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix} \quad \lambda_1 = 1 \quad \lambda_2 = .7$$

$$A - I = \begin{bmatrix} -.1 & .2 \\ .1 & -.2 \end{bmatrix} \quad x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A - 0.7I = \begin{bmatrix} .2 & .2 \\ .1 & .1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$u_k = c_1 \cdot 1^k \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \cdot (0.7)^k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$c_1 = \frac{50}{3} \qquad c_2 = 10 - \frac{100}{3} = -\frac{70}{3}$$